# CAPSTONE PROJECT REVIEW PROGRESS RECORD

**Review:** 03
**Date of Review:** 06.11.2026
**Project Title:** Sensitivity-Guided Layer-Wise Approximate Multiplier Mapping for FPGA-Based CNN Accelerators
**Cohort:** Dr. V. Vaishnavi
**Project Guide:** Dr. Arun Nimmala
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## Brief Description of the Project

Convolutional Neural Network (CNN) edge accelerators are constrained by the high area and power dissipation of Multiply-Accumulate (MAC) datapaths, where 8x8 exact multipliers dominate silicon cost (68 LUTs on Xilinx Zynq-7020). While approximate computing trades controlled arithmetic error for hardware savings, uniform approximation causes severe accuracy degradation or negligible gains. Existing layer-wise mapping methods rely on computationally prohibitive evolutionary search (ALWANN: 7.5 CPU-days) or GPU retraining loops (TRAM). This project proposes a lightweight, single-pass sensitivity metric computed in 0.2 seconds combining static weight variance and dynamic loss gradients via a 256-sample calibration probe without retraining. Layers are ranked such that error-vulnerable early layers receive exact multipliers while noise-resilient deep layers receive aggressive approximate multipliers. The heterogeneous design is mapped onto synthesizable Verilog Processing Elements (PEs) and deployed on a PYNQ-Z2 FPGA board via AXI4-Stream DMA, achieving up to 32.5% multiplier area reduction while preserving baseline classification accuracy.

## Progress Achieved up to This Review

- **Pareto Knee Multi-Objective Benchmark:** Executed 256-configuration NSGA-II multi-objective exploration; proved the single-pass sensitivity assignment lands on the Pareto knee in 0.2s (>3,000,000x faster than ALWANN evolutionary search).
- **Synthesizable Accelerator Datapath:** Implemented complete Verilog RTL (axm_8x8, mac_unit, conv_pe, layer_controller, cnn_accelerator_top) with AXI4-Stream interfaces; verified via 262,144-cycle testbench with 100% vector match.
- **Vivado Block Design Integration:** Created automated vivado_block_design.tcl synthesizing and packaging custom IP, integrating ARM Cortex-A9 PS, AXI DMA engines (MM2S & S2MM), and interconnect at 100 MHz.
- **PYNQ-Z2 Deployment Stack:** Exported quantized INT8 weights/biases (weights_int8.npy), developed Python runtime driver (pynq_inference.py), and created interactive demonstration Jupyter notebook (pynq_cnn_demo.ipynb).
- **Academic Dissemination & GitHub:** Authored 6-page IEEE conference paper, generated 7 publication-quality figures, created 20-slide presentation with speaker notes, and published full repository to GitHub.

**Signature of Project Guide:** _____ **Date:** _____